Cloud Computing
LAB 7

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1. Aim

To implement and understand Block Storage (Ceph) and Object Storage (MinIO), and analyze their behavior through experiments.

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2. Procedure with Screenshots (IN ORDER)

Step 1: Folder Setup

Created a folder with SRN and extracted lab files.

```
 => resolving provenance for metadata file                    0.0s
[+] up 2/2
 ✔Image pes1ug23am343_lab7-webapp        Built                 6.1s
 ✔Container pes1ug23am343_lab7-webapp-1 Recreated              0.8s
Attaching to minio-1, webapp-1
minio-1   | MinIO Object Storage Server
minio-1   | Copyright: 2015-2026 MinIO, Inc.
minio-1   | License: GNU AGPLv3 - https://www.gnu.org/license
s/agpl-3.0.html
minio-1   | Version: RELEASE.2025-09-07T16-13-09Z (go1.24.6 l
inux/amd64)
minio-1   |
minio-1   | API: http://172.19.0.2:9000  http://127.0.0.1:900
0
minio-1   | WebUI: http://172.19.0.2:9001 http://127.0.0.1:90
01
minio-1   |
minio-1   | Docs: https://docs.min.io
webapp-1  | Starting Flask app on http://0.0.0.0:5000
webapp-1  |  * Serving Flask app 'app'
webapp-1  |  * Debug mode: off
webapp-1  | WARNING: This is a development server. Do not us
e it in a production deployment. Use a production WSGI serve
r instead.
webapp-1  |  * Running on all addresses (0.0.0.0)
webapp-1  |  * Running on http://127.0.0.1:5000
webapp-1  |  * Running on http://172.19.0.3:5000
webapp-1  | Press CTRL+C to quit
webapp-1  | 172.19.0.1 - - [24/Mar/2026 04:37:43] "GET / HTT
P/1.1" 200 -
webapp-1  | 172.19.0.1 - - [24/Mar/2026 04:37:44] "GET /favi
con.ico HTTP/1.1" 404 -

v View in Docker Desktop   o View Config   w Enable Watch
d Detach
```

Step 2: Modify Bucket Name in app.py

Updated bucket name to SRN.

Step 3: Run Docker

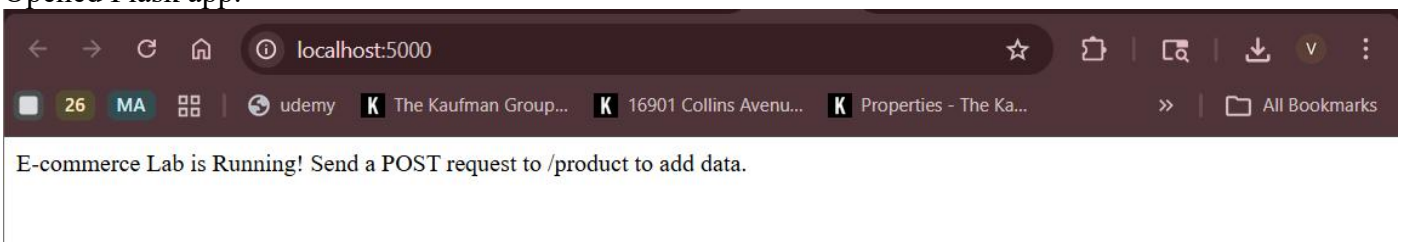
docker-compose up --build

```
C:\Users\acer>docker ps
CONTAINER ID   IMAGE                        COMMAND                  CREATED          STATUS          PORTS
                                                NAMES
6b1002eac7da   pes1ug23am343_lab7-webapp   "python app.py"          17 minutes ago   Up 17 minutes   0.0.0.0:5000->5000/
tcp, [::]:5000->5000/tcp                        pes1ug23am343_lab7-webapp-1
9602e26335e4   minio/minio                 "/usr/bin/docker-ent…"   41 minutes ago   Up 17 minutes   0.0.0.0:9000-9001->
9000-9001/tcp, [::]:9000-9001->9000-9001/tcp   pes1ug23am343_lab7-minio-1

C:\Users\acer>
```

Step 4: Flask Application Running

Opened Flask app.

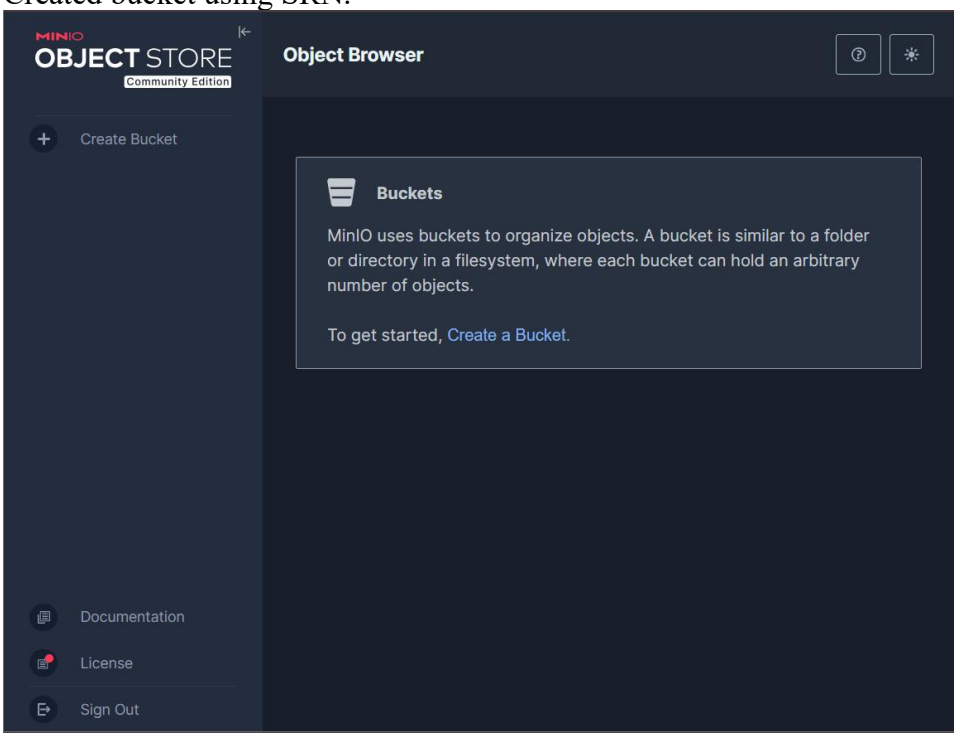


Step 5: MinIO Dashboard Login

Opened MinIO dashboard and logged in.

Step 6: Bucket Creation

Created bucket using SRN.



Step 7: Upload Image using curl

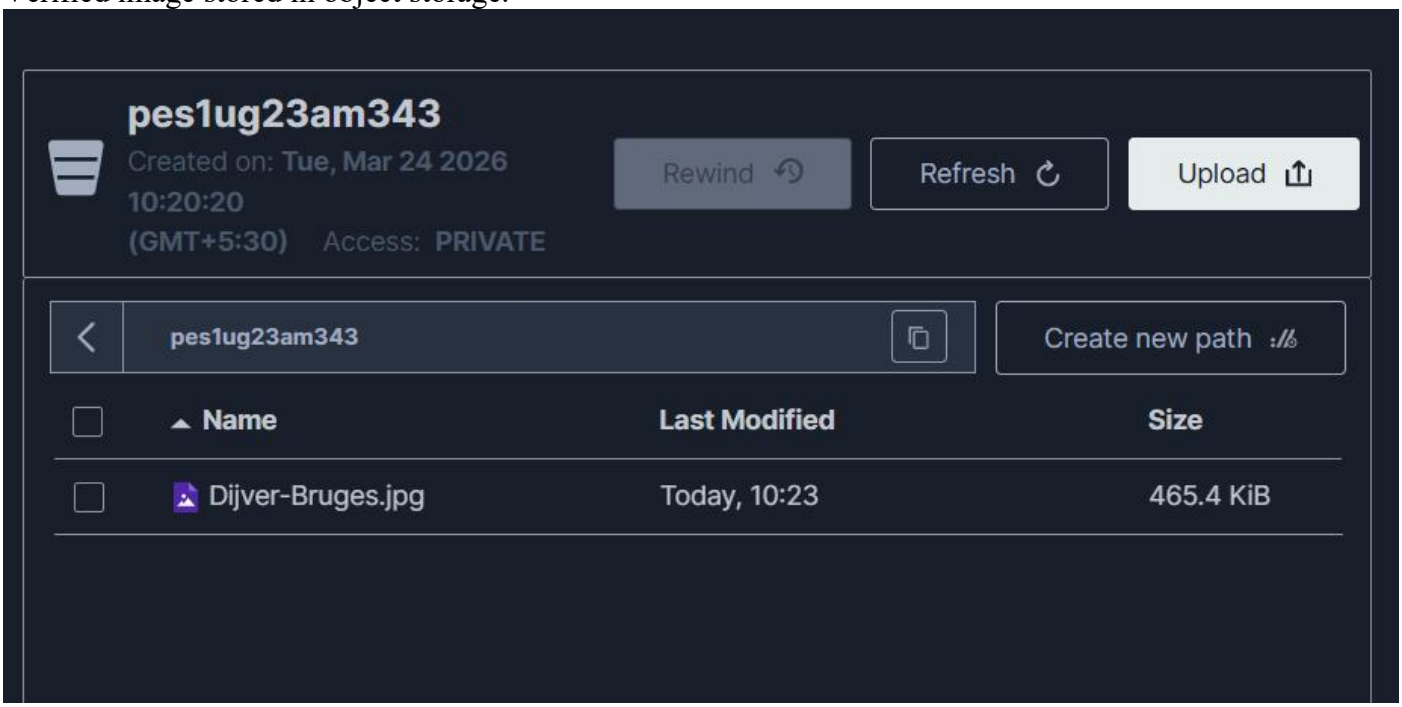
curl command to upload image.

```
C:\Users\acer>curl.exe -X POST -F "name=Phone" -F "price=500" -F "image=@\"C:\Users\acer\OneDrive\Pictures\Dijver-Bruges
.jpg\"" http://localhost:5000/product
{"msg":"Structured data in Block, Image in Object","status":"success"}

C:\Users\acer>
```

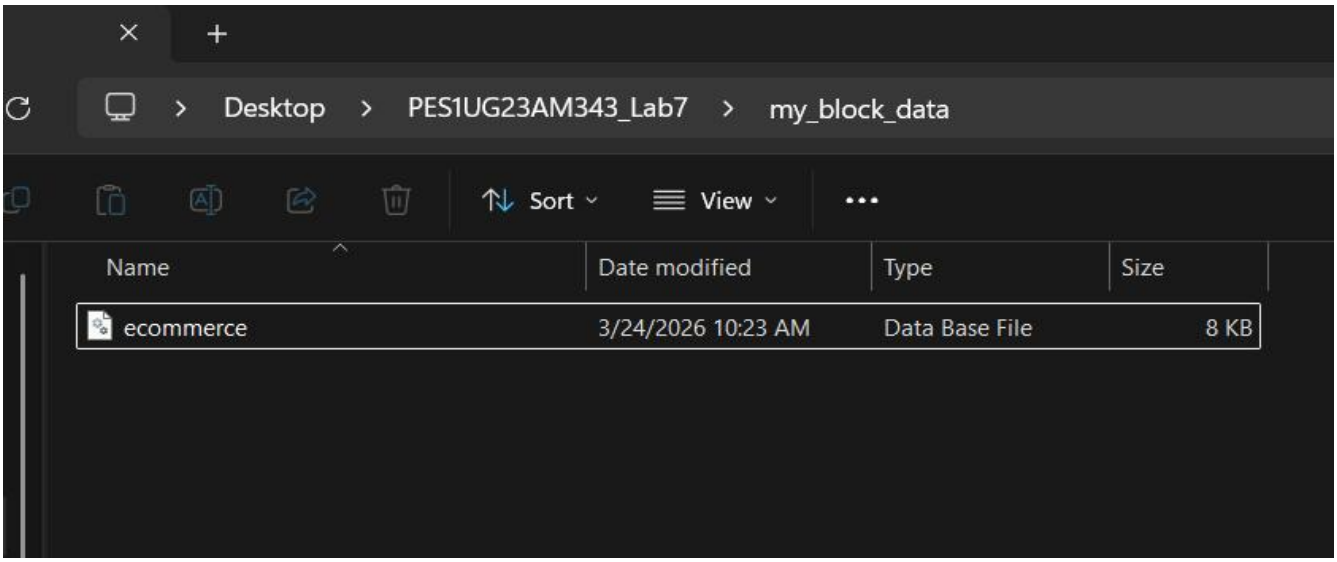
Step 8: Image Stored in MinIO

Verified image stored in object storage.



Step 9: Block Storage Verification

Checked local folder for database



Step 10: Database Query Output

docker command to view stored data.

```
C:\Users\acer>docker exec -it pes1ug23am343_lab7-webapp-1 python -c "import sqlite3; conn = sqlite3.connect('/mnt/block_
volume/ecommerce.db'); cursor = conn.cursor(); cursor.execute('SELECT * FROM product'); print(cursor.fetchall()); conn.c
lose()"
[(1, 'Phone', 500.0, 'Dijver-Bruges.jpg')]
```

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3. Experiment 1: Access Method

Observation:

- Database file is visible locally
- Image is not present locally
- Image exists only in MinIO

Conclusion:

Block storage behaves like a disk, while object storage is accessed via API.

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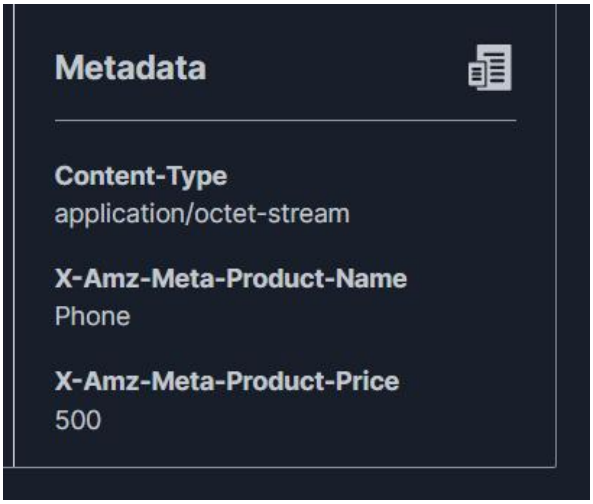
4. Experiment 2: Metadata

Added metadata in app.py:

```
file_metadata = {
    "x-amz-meta-product-name": str(name),
    "x-amz-meta-product-price": str(price)
}
```

Uploaded image again and verified metadata.

```
C:\Users\acer>curl.exe -X POST -F "name=Phone" -F "price=500
" -F "image=@\"C:\Users\acer\OneDrive\Pictures\Dijver-Bruges
.jpg\"" http://localhost:5000/product
{"msg":"Structured data in Block, Image in Object","status":
"success"}
```



Observation:

Metadata is visible without opening the file.

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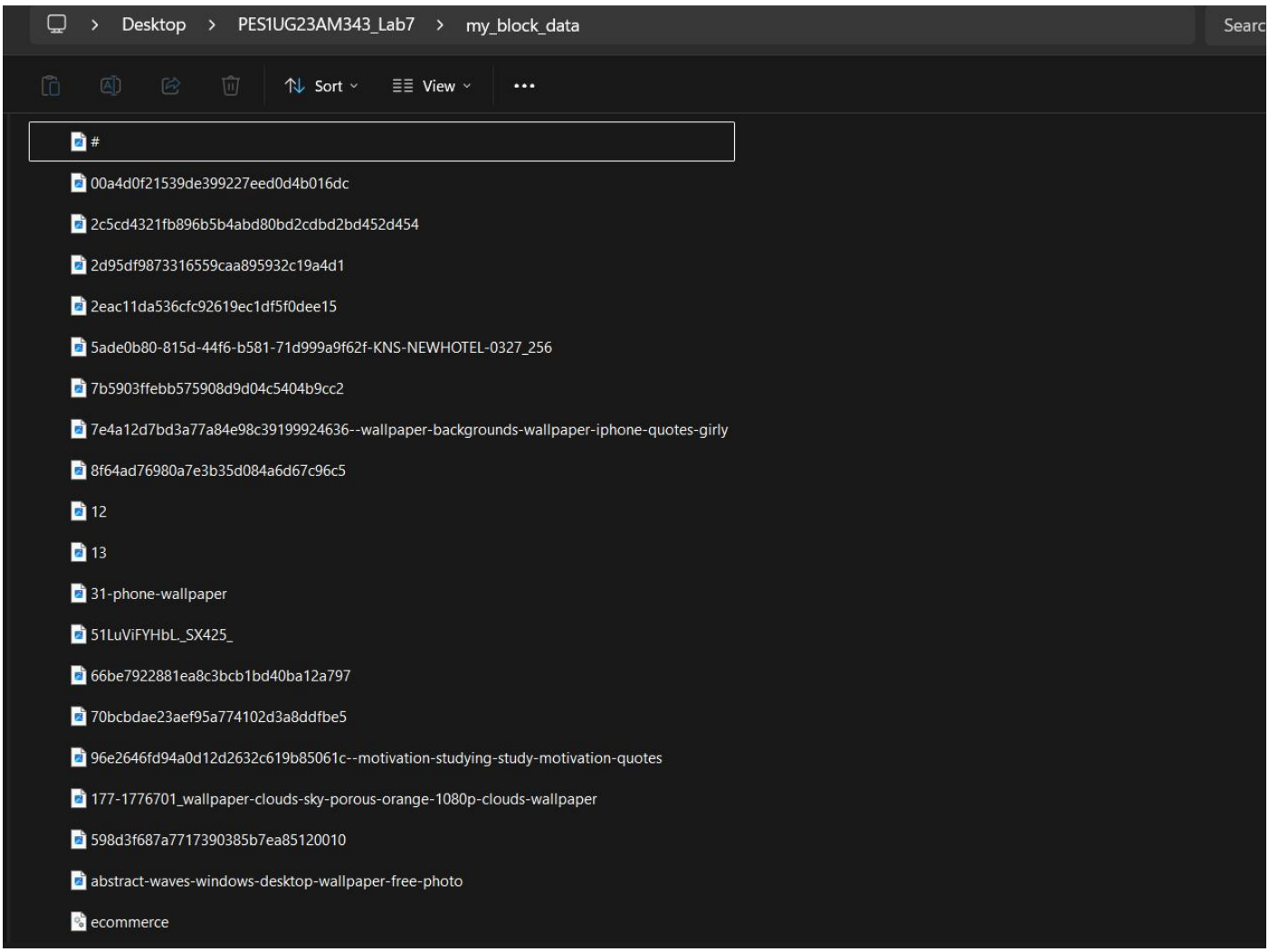
5. Experiment 3: Scaling

Filled block storage folder with files.

Observation:

- Block storage shows limitation (disk full scenario)
- MinIO continues to work



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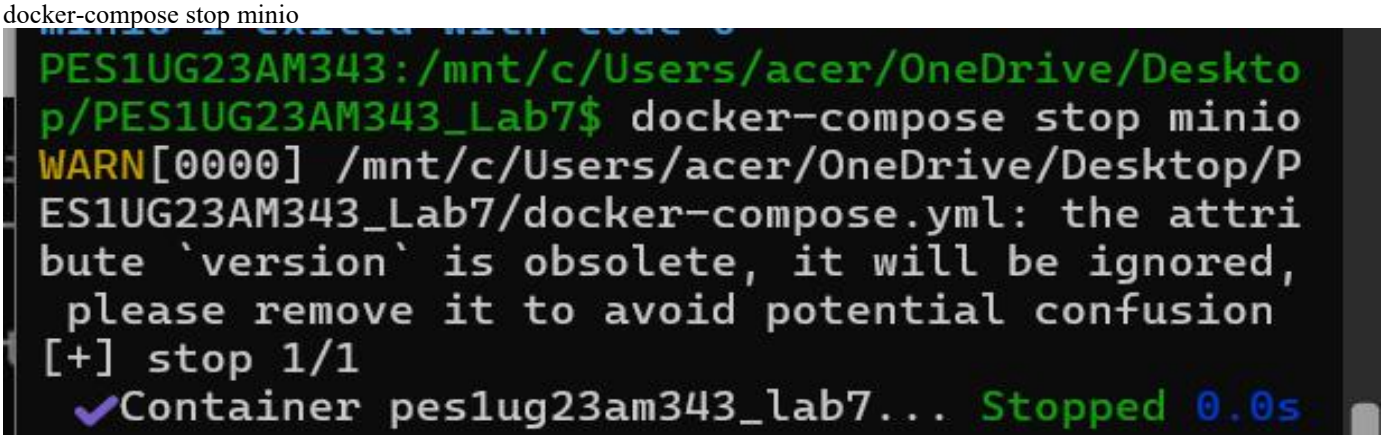
6. Experiment 4: Failure Testing

Step 1: Stop MinIO

docker-compose stop minio



```
C:\Users\acer>curl.exe -X POST -F "name=Phone" -F "price=500
" -F "image=@\"C:\Users\acer\OneDrive\Pictures\Dijver-Bruges
.jpg\"" http://localhost:5000/product
curl: (7) Failed to connect to localhost port 5000 after 224
8 ms: Could not connect to server
```

Step 2: Stop Webapp

docker-compose stop webapp

```
p/PES1UG23AM343_Lab7$ docker-compose stop webap
p
WARN[0000] /mnt/c/Users/acer/OneDrive/Desktop/P
ES1UG23AM343_Lab7/docker-compose.yml: the attri
bute `version` is obsolete, it will be ignored,
 please remove it to avoid potential confusion
[+] stop 1/1
 ✔Container pes1ug23am343_lab7... Stopped 2.1s
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Deskto
p/PES1UG23AM343_Lab7$
```

Observation:

- When MinIO is down->image upload fails
- When webapp is down-> data persists

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7. Conclusion

This lab demonstrated the difference between block storage and object storage. Block storage is used for structured data like databases, while object storage is used for unstructured data like images. Object storage supports metadata and scalability, whereas block storage has size limitations.